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Permian to Early Triassic magnetostratigraphy from the Central European Basin in Poland: Implications on regional and worldwide correlations

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Abstract

Permian–Early Triassic rocks sampled in 15 outcrops (93 hand samples) and 11 unoriented boreholes (630 hand samples) from the Polish part of the Central European Basin were studied paleomagnetically. In spite of extensive remagnetization, affecting 50% of samples from the boreholes, a magnetic polarity sequence was compiled. The polarity pattern is not only common for many sections from the investigated area but also convergent with the global synthetic magnetostratigraphic scale. The Lower Rotliegend (Early Permian) rocks contain a reversed polarity record, while the Upper Rotliegend (Late Permian)–Early Triassic sediments are characterized by mixed polarity magnetization. Within the Buntsandstein part of that mixed megazone 14 magnetic zones were found. Buntsandstein sedimentation started with a normal polarity zone, which may be compared with that of Early Griesbachian defined in several places in the world. The polarity record noted in the Middle Buntsandstein sediments could indicate significant stratigraphic gaps or condensations near the Middle Buntsandstein and Röt boundary. The comparison of the Upper Buntsandstein polarity with the Early–Middle Triassic magnetostratigraphic data from other basins suggests that the majority or even all the Röt sediments should be included in the Anisian. © 1997 Elsevier Science B.V.

Keywords: Poland; magnetostratigraphy; Bunter; Lower Triassic; Anisian

1. Introduction

Magnetostratigraphy can be used as a tool for chronostratigraphic correlations between rocks from different environments including those with a poor biotic documentation. Permian strata have been magnetostratigraphically studied in many places in the world. Among them, the end of long reversed polarity Kiaman superchron [1] was indicated. This point (i.e. Illawarra reversal) was proposed in the uppermost Permian of Australia [2], and was reported in the Tatarian of the Russian Platform [3], in the Ochoan [4] or Guadalupian [53] of North America and in the Upper Rotliegend of Western Europe [5,6]. Results of studies on the Permian–Triassic transition sequences in the Arctic area [7,8], China and Pakistan sections [9–11] as well as in Siberian traps [12] show that the uppermost Permian rocks are mixed magnetized while the lowermost Griesbachian

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strata have recorded a relatively long normal polarity zone. Preliminary studies of Zechstein–Buntsandstein transition beds from Poland have revealed the same magnetostratigraphic behaviour as that observed in other Permian–Triassic transition beds [13].

In recent years, progress has also been made in calibration of the sequence of the Triassic reversed and normal polarity magnetozones against a detailed biostratigraphic zonation, both in the Boreal and Tethyan areas [7,8,14]. The Early Triassic rocks from the Central European Basin (CEB) were not studied magnetostratigraphically, except for preliminary investigations carried out in the German southernmost margin of the CEB about 30 years ago [5,15]. Since that time, however, the Buntsandstein stratigraphy of the CEB has been remarkably developed. Palynostratigraphy and sequence stratigraphy have been successfully applied not only for internal (basin) but also for global correlations (e.g. [16,17]).

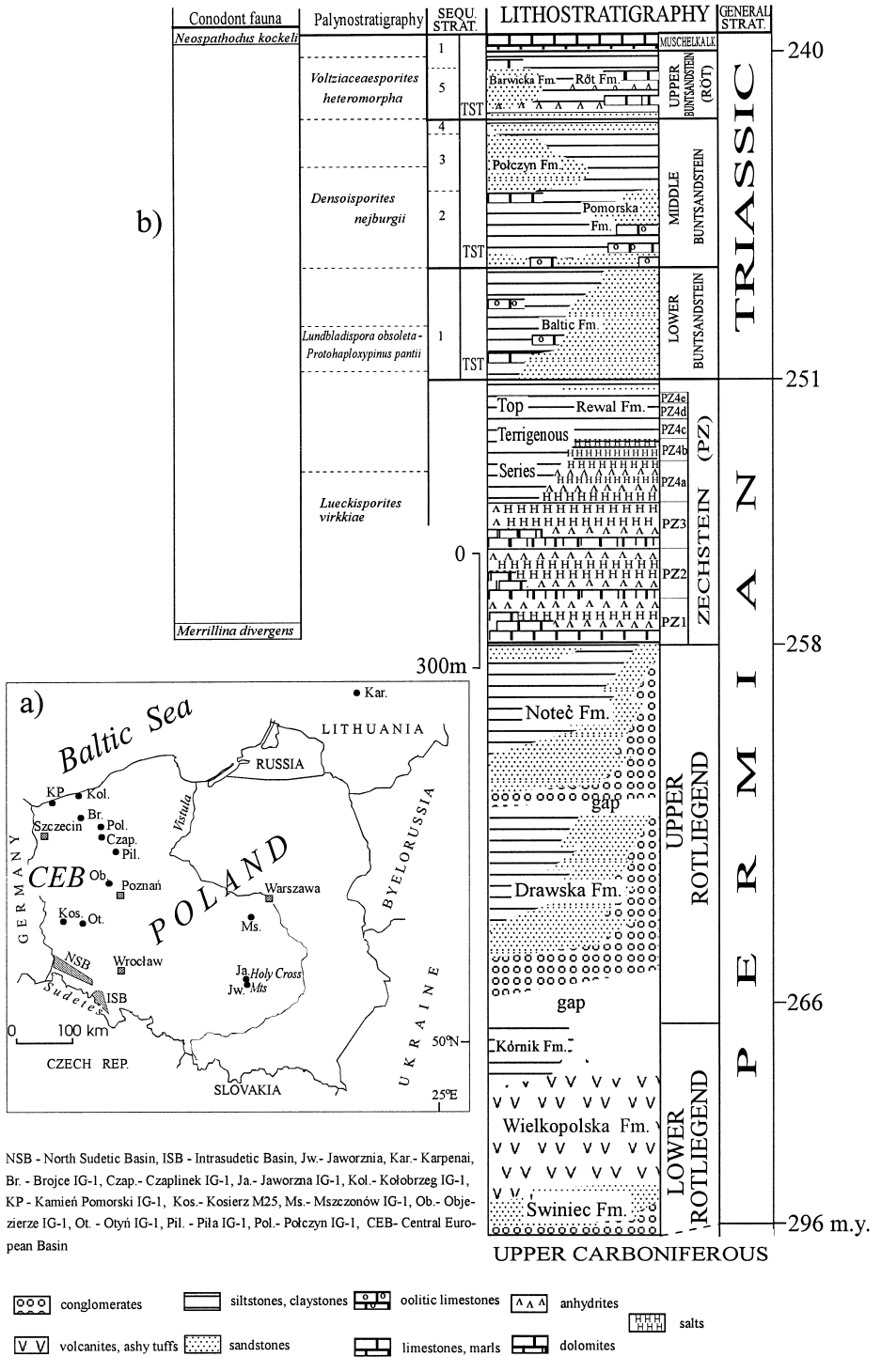
The aim of this paper is to complete the magnetostratigraphic scale for the Permian and Early Triassic sequences from the Polish part of the CEB. Paleomagnetic studies should answer the question to what degree the lithostratigraphic boundaries defined between Permian and Early Triassic formations in Central Europe are related to stratigraphic units in the Boreal and Tethyan realms.

2. Permian–Early Triassic stratigraphy in Poland

The division and correlation of the Lower Permian rocks in Poland is based on the results of lithostratigraphic analysis. The Lower Rotliegend ('Autunian') succession consists mainly of volcanogenic rocks which are underlain and covered by a less thick clastic series (Fig. 1). A large (not precisely defined) stratigraphic gap between Lower and Upper Rotliegend has been postulated [18]. Such a gap was also indicated in Rotliegend profiles of the German part of the CEB [6] where the Lower Rotliegend rocks were deposited between 300 and 285 Ma [19]. The Upper Rotliegend sediments in Poland belong to the 'red beds' association developed mainly in river and desert environments [20]. Two megacycles of sedimentation beginning with conglomerates were distinguished in the central part of the Polish Basin [18]. In the Upper Rotliegend sediments from the German part of the CEB, the Illawarra Reversal, ending the Kiaman reversed superchron, of a maximum estimated age of 265 Ma, was found [5,6,19].

The lithostratigraphic division based on the division of cyclothems according to the cyclic sedimentation of evaporites has been used for correlations within the Polish Zechstein Basin, which represents the eastern part of the extensive CEB. The three oldest cyclothems: PZ1, PZ2, and PZ3 are built of carbonate–evaporite sequences which formed during the transgressive–regressive cycles of the Zechstein sea. Five subcyclothems make up a terrigenous–evaporitic sequence that formed as a result of cyclic climatic changes in cyclothem PZ4 [21]. Series of mostly red clastic rocks (the so called Top Terrigenous Series [21]), which overlie the cyclothem PZ4 and replace it in the marginal parts of the Polish Zechstein Basin, are also considered as belonging to the Zechstein. In the bottom part of these sediments the Late Permian palynomorph *Lueckisporites virkkiae* was found [22]. The lowermost beds of the Polish Zechstein contain the conodont *Merrillina divergens* known also from the western part of the CEB [23]. The palynological correlation, verified by carbon isotope analyses, sedimentological investigations and preliminary magnetostratigraphic studies, seems to indicate that the Zechstein–Buntsandstein boundary in the Polish profile is very close to the Permian–Triassic boundary as defined in the Arctic area [13,24–27]. An increased value of ^{13}C isotope in the Zechstein carbonates has been observed in the Late Permian rocks from the other basins, suggesting that the Zechstein is younger than Capitanian and should be correlated with the Ochoan stage [25]. Menning correlates the Zechstein with the main part

Fig. 1. (a) Location map of boreholes, quarries and local sedimentary basins where Permian–Early Triassic rocks were sampled. (b) Composite lithostratigraphic column of Permian and Early Triassic sequences from the Polish part of the Central European Basin (after [18,21,28], simplified). Sequence stratigraphic boundaries were adopted from [17] (TST = transgressive system tracks). Ages after [19].


b)
Conodont fauna
Palynostratigraphy
SEQU. STRAT.
LITHOSTRATIGRAPHY
GENERAL STRAT.
Neospathodus kockeli
Voltziaceaesporites heteromorpha
Densoisporites nejburgii
Lundbladispora obsoleta - Protohaploxypinus pantii
Lueckisporites virkkiae
Merrillina divergens
MUSCHELKALK
Barwicka Fm.
Röt Fm.
UPPER BUNTSANDSTEIN (RÖT)
Połczyn Fm.
Pomorska Fm.
MIDDLE BUNTSANDSTEIN
Baltic Fm.
LOWER BUNTSANDSTEIN
TRIASSIC
TST
Top
Rewal Fm.
Terrigenous Series
PZ4e
PZ4d
PZ4c
PZ4b
PZ4a
PZ3
PZ2
PZ1
ZECHSTEIN (PZ)
0
300m
Noteć Fm.
gap
UPPER ROTLIEGEND
Drawska Fm.
gap
Kórnik Fm.
Wielkopolska Fm.
Swiniec Fm.
LOWER ROTLIEGEND
PERMIAN
240
251
258
266
296 m.y.
UPPER CARBONIFEROUS
a)
Baltic Sea
LITHUANIA
RUSSIA
Kar.
KP
Kol.
Br.
Szczecin
Pol.
Czap.
Pil.
Vistula
CEB
Ob.
Poznań
Warszawa
POLAND
BYELORUSSIA
GERMANY
Kos.
Ot.
Ms.
NSB
Wrocław
Ja.
Holy Cross Mts
Jw.
Sudetes
ISB
0 100 km
CZECH REP.
SLOVAKIA
UKRAINE
50°N
25°E
NSB - North Sudetic Basin, ISB - Intrasudetic Basin, Jw.- Jaworznia, Kar.- Karpenai, Br. - Brojce IG-1, Czap.- Czaplinek IG-1, Ja.- Jaworzna IG-1, Kol.- Kołobrzeg IG-1, KP - Kamień Pomorski IG-1, Kos.- Kosierz M25, Ms.- Mszczonów IG-1, Ob.- Objezierze IG-1, Ot. - Otyń IG-1, Pil. - Piła IG-1, Pol.- Połczyn IG-1, CEB- Central European Basin
conglomerates
siltstones, claystones
oolitic limestones
anhydrites
salts
volcanites, ashy tuffs
sandstones
limestones, marls
dolomites

of the Ochoan and the Upper Tatarian [53] and estimates its duration to 7 Ma [19].

The Buntsandstein Series in Poland developed mainly as red clastic beds and form the lower part of the Germanic Triassic. It was deposited during the Late Permian to Early Anisian in a fluvio-lacustrine environment, with marine influences [17,28,29]. The Buntsandstein is traditionally subdivided into the fine-grained Lower Buntsandstein Series, comprising several oolite beds, the Middle Buntsandstein, composed mainly of sandstones alternating with claystones, and the Upper Buntsandstein (Röt) represented by evaporites and claystones occasionally replaced by sandstones. Sequence stratigraphy has been applied only to deposits in the German and Dutch parts of the Buntsandstein Basin [17,29]. Because of the very strong similarity of the Buntsandstein from Germany and Poland, elements of that stratigraphy were directly transferred into the Polish synthetic profile (Fig. 1).

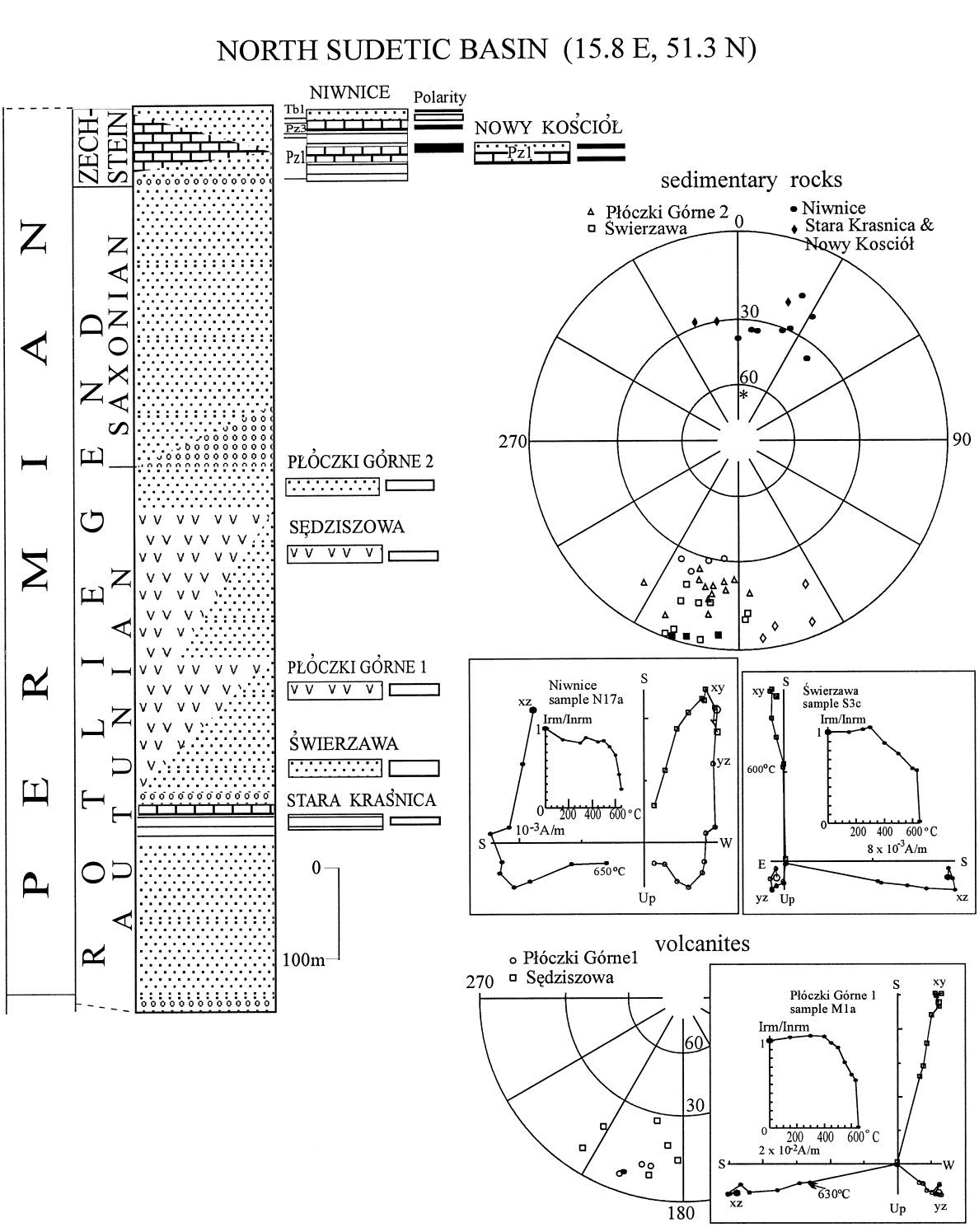
Buntsandstein sedimentation started in the Polish area with marine transgressive clastic sequences [27] containing microspores *Lundbladispora obsoleta–Protohaploxypinus panti* of Griesbachian age [30]. The upper part of the Middle Buntsandstein (Połczyn Formation) was deposited in fluvial environments [28]. In that part of the Polish Buntsandstein the occurrence of hiatuses is very likely. The chronostratigraphic position of the Upper Buntsandstein (Röt) sediments is somewhat controversial. Some authors have located the whole or the majority of the Röt in the Anisian [17,31–33]. Others have connected only the upper part of it with the Anisian [19,28,34]. Conodont faunas of Pelsonian age were found in the lower (but not the lowermost) part of the Polish Muschelkalk [35]. On the other hand, the presence of *Volziaceasporites heteromorpha* and *Microcachrydites fastidiosus* in Röt sediments from Poland indicates that at least the upper part of the Röt corresponds to the Anisian [33].

3. Sampling localities and paleomagnetic procedure

A total of 630 Permian–Early Triassic samples were collected for paleomagnetic investigations from 11 vertical boreholes situated in different areas of the Polish part of the CEB (Fig. 1). These samples were only oriented top–bottom. The Permian rocks were also sampled in the outcrops located in the Intrasudetic and North Sudetic basins, and in the Holy Cross Mts. (Jaworznia quarry). Moreover, Late Permian rocks were sampled in the northern part of Lithuania (Karpenai quarry). Beds of the rocks studied were characterized by a very small dip that, in the boreholes and Lithuanian outcrop, did not exceed few degrees. In the Sudetes and the Holy Cross Mts. it was usually less than 20°. A set from the outcrops contained 93 hand samples which were taken from relatively narrow stratigraphic intervals. Therefore data from the outcrops were used mainly as control points for the data set from the boreholes.

Several core specimens 2.2 cm in diameter and length were drilled from each sample (i.e. from a fragment of drill core, bed or lava flow). The natural remanent magnetization (NRM) of specimens was measured using a JR-5 spinner magnetometer. Each specimen has been subjected to stepwise demagnetization in a μ-metal shielded oven, which reduced the ambient field close to a few nT. Magnetic susceptibility was monitored using a KLY-2 susceptibility bridge. Demagnetization and measurements were carried out in a cage, reducing the ambient field by about 95%. Principal component analysis, as presented by Kirschvink [36], was used to calculate the

Fig. 2. Synthetic lithostratigraphic column of the Permian succession in the North Sudetic Basin (after [48]), and the results of paleomagnetic investigations (stereographic projection of characteristic high-temperature directions, orthogonal vector plots and intensity decay curves prepared for representative samples; Inrm = intensity of natural remanent magnetization, Irm = intensity of remanent magnetization after thermal treatment, xy,xz,yz = planes of orthogonal projection; an asterisk on the stereonet means the local present-day field direction). On the right of the lithostratigraphic column, the profiles of investigated outcrops and magnetic polarities obtained are presented (black belt = normal polarity record, white belt = reverse polarity record). The age of the investigated sections is the same as the age of neighbouring parts of synthetic profile. Lithological symbols as in Fig. 1.


NORTH SUDETIC BASIN (15.8 E, 51.3 N)
NIWNICE
Polarity
NOWY KOŚCIÓŁ
Tb1
Pz3
Pz1
ZECH-STEIN
SAXONIAN
ROTLIEGEND
AUTUNIAN
PERMIAN
PŁÓCZKI GÓRNE 2
SĘDZISZOWA
PŁÓCZKI GÓRNE 1
ŚWIERZAWA
STARA KRAŚNICA
0
100m
sedimentary rocks
Płóczki Górne 2
Świerzawa
Niwnice
Stara Krasnica &
Nowy Kosciół
0
30
60
270
90
Niwnice
sample N17a
Irm/Inrm
xz
xy
yz
S
W
Up
650°C
10-3A/m
Świerzawa
sample S3c
600°C
8 x 10-3A/m
E
volcanites
Płóczki Górne1
Sędziszowa
270
60
30
180
Płóczki Górne 1
sample M1a
2 x 10-2A/m
630°C

components of NRM and their unblocking temperature spectra. Thermomagnetic analysis and analysis of isothermal remanent magnetization (IRM) acquisition were used to determine the nature of magnetic carriers.

Results from each borehole sample were not treated in the same way. They were divided into three categories of reliability. In the first category the characteristic direction was calculated by line fitting [36]. The mean direction, calculated from two or more fitted lines, had to reveal internal homogeneity ($\alpha_{95} < 25°$). The second category of paleodirection was based on the samples containing also at least two specimens with good internal homogeneity but the characteristic directions were calculated there as a stable end point. Results from samples containing

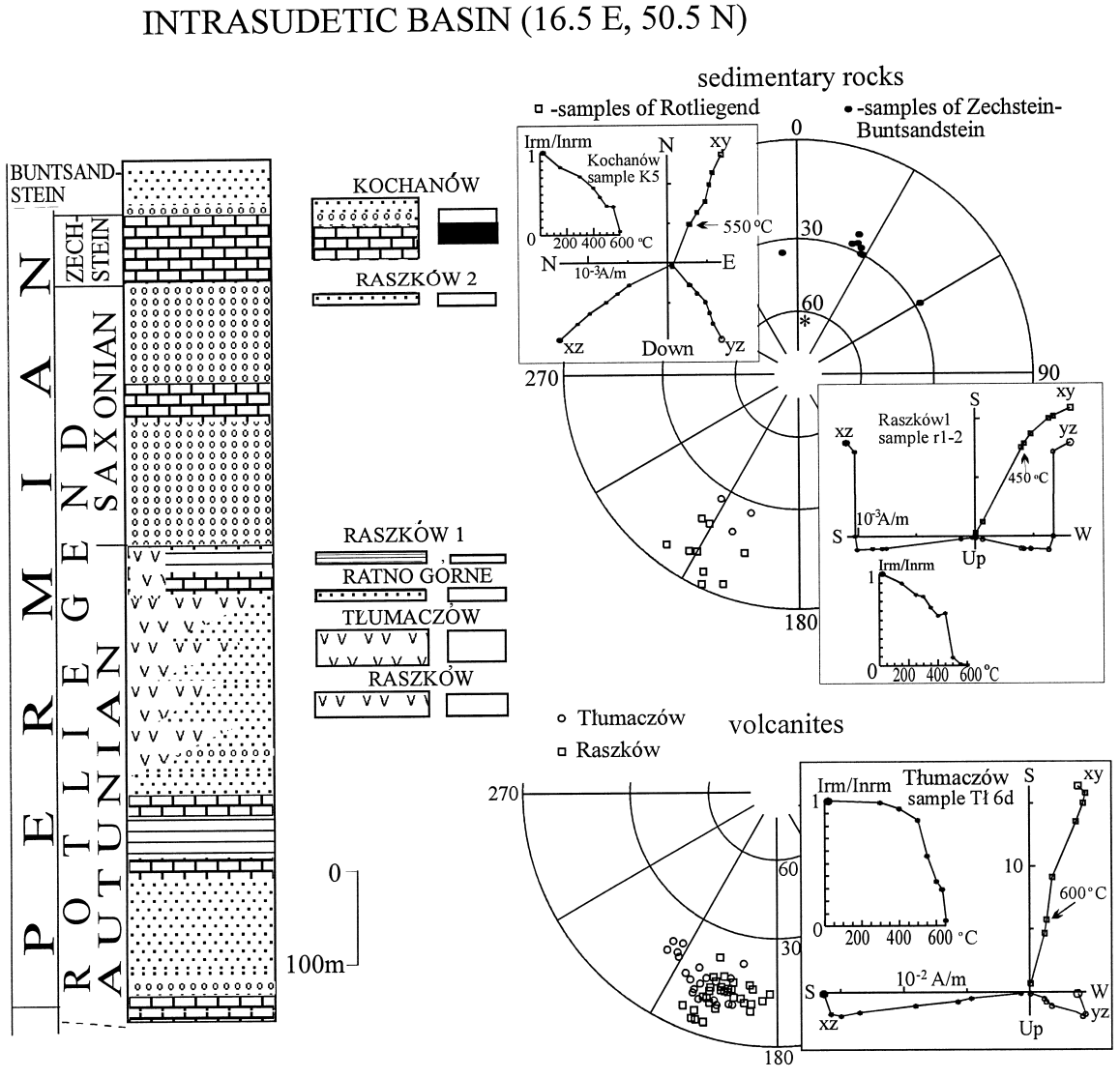
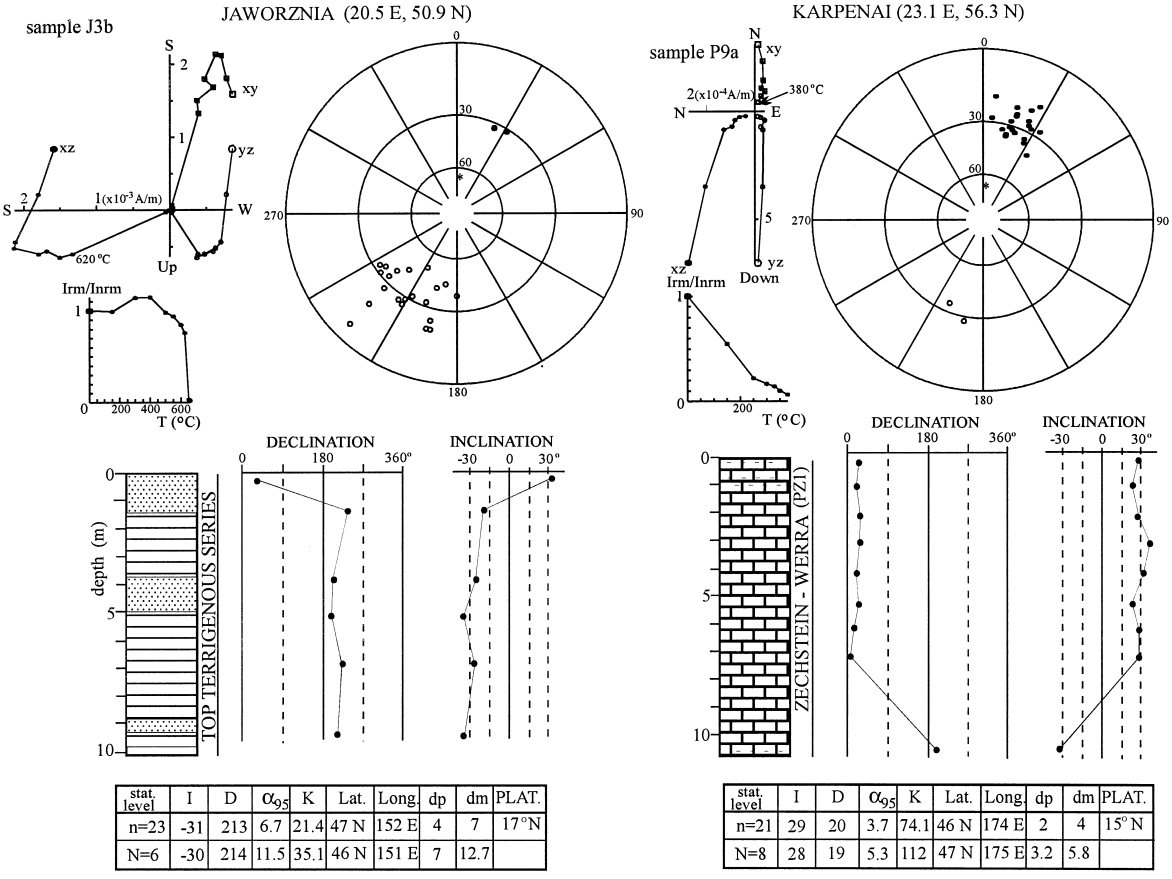


Fig. 3. Synthetic lithostratigraphic column of Permian succession from the Intrasudetic Basin (after [48]) and the results of paleomagnetic investigations. For explanation see Fig. 2.

Table 1
Summary of statistics of Permian paleodirections isolated in the North Sudetic and Intrasudetic basins

Area	Age	Rocks	N/n	D	I	α_{95}	K	Lat.	Long.	dp	dm	Pol.
North Sudetic Basin	Zechstein	Sedimentary	10/14	17	29	6.4	39	51°N	168°E	3.9	7.1	Mix
	Autunian	Sedimentary	8/29	188	−12	5.1	28.2	44°N	184°E	2.6	5.2	R
	Autunian	Volcanites	6/10	196	−11	8.1	36.4	42°N	174°E	4.2	8.2	R
Intrasudetic Basin	Zechstein	Sedimentary	8/11	23	29	4.2	148.7	51°N	160°E	2.4	4.4	Mix
	Autunian	Sedimentary	4/6	208	−6	6.8	99.1	37°N	160°E	3.4	6.8	R
	Autunian	Volcanites	13/54	198	−12	3.2	116	43°N	172°E	1.5	2.8	R
Sudetes (summary statistics)	Zechstein		18/25	21	29	4.9	35.3	50°N	163°E	3.0	5.4	86%N 14%R
	Autunian		8 [a]/99	199	−11	5.0	121.8	42°N	171°E	2.6	5.1	R

N = number of beds or lava flows; n = number of specimens; D = declination; I = inclination; K = Fisher statistics parameters [52]; Lat. = latitude of north paleomagnetic pole; Long. = longitude of north paleomagnetic pole; dp = error of the distance between site and paleopole; dm = paleodeclination error; Pol. = polarity (Mix = mixed, N = normal, R = reversed).
[a] Number of localities.



stat. level	I	D	α_{95}	K	Lat.	Long.	dp	dm	PLAT.
n=23	-31	213	6.7	21.4	47 N	152 E	4	7	17°N
N=6	-30	214	11.5	35.1	46 N	151 E	7	12.7	

stat. level	I	D	α_{95}	K	Lat.	Long.	dp	dm	PLAT.
n=21	29	20	3.7	74.1	46 N	174 E	2	4	15° N
N=8	28	19	5.3	112	47 N	175 E	3.2	5.8	

Fig. 4. Results of paleomagnetic investigations (orthogonal plots, intensity decay curves, stereographic projections of characteristic directions, declination and inclination graphs close by lithological columns, and tables with summary statistics of isolated paleodirections) of Late Zechstein red clastic sediments from Jaworznia (Holy Cross Mts.) and Early Zechstein carbonates from Karpenai (Lithuania). PLAT = paleolatitude, other explanations as in Fig. 2 and Table 1.

only one specimen, where the characteristic direction was calculated either as the fitted line or the stable end direction, were classified to the third category of reliability. In each category the characteristic inclinations had to correspond to the expected Permian–Early Triassic inclinations. It is generally accepted that the area of investigation belonged to stable Europe at least from the Permian onwards. Early Permian paleomagnetic results from stable Europe [37] give reversed polarity directions for Poland displaying declination values which usually lie between 190° and 210°, and inclination values of −10° to −25°. Late Permian–Early Triassic stable European results give directions for Poland with the same range of declinations and inclinations of reversed polarity directions, which lie within the range −25° to −35°.

4. Results of paleomagnetic analysis

4.1. Samples from outcrops

4.1.1. North Sudetic Basin

Permian sedimentary and volcanic rocks were studied from seven localities. One distinct component of NRM, with blocking temperatures exceeding 600°C, occurs in Lower Rotliegend ('Autunian') volcanites and red clastic rocks (Fig. 2). This component is only of reversed polarity in all outcrops containing different Autunian horizons. In the Zechstein carbonates and clastic sediments from the North Sudetic Basin [38] the structure of NRM is more complex. Components with shallow inclination occur only in a part of the samples and were isolated at temperatures above 400°C. In most Zechstein samples, shallow inclination directions are of normal polarity. Only samples from the upper part of Zechstein were reversely magnetized.

4.1.2. Intrasudetic Basin

Autunian volcanites, sampled in Raszków and Tłumaczów quarries, were characterized by the same simple NRM structure as the volcanites from the North Sudetic Basin. A comparable reversed component was demagnetized at temperatures above 600°C (Fig. 3). Maximum unblocking temperatures of shallow inclination components below 600°C and the frequent presence of a steep low temperature component were characteristic for grey Autunian mudstones and carbonates attributed to the Zechstein [39]. Most of the samples from these rocks were normally magnetized (Table 1).

4.1.3. Holy Cross Mts., Jaworznia quarry

The NRM of Late Zechstein red sandstones and mudstones consists of two components. The first component, with a steep inclination, was removed at temperatures up to 300°C. The second, shallow inclination, component was completely demagnetized at temperatures of 630–650°C. With its mean inclination of −30°, it is located in the third quarter of the hemisphere (Fig. 4). Only a high-temperature direction of one sample from the uppermost part of section is oriented antiparallel.

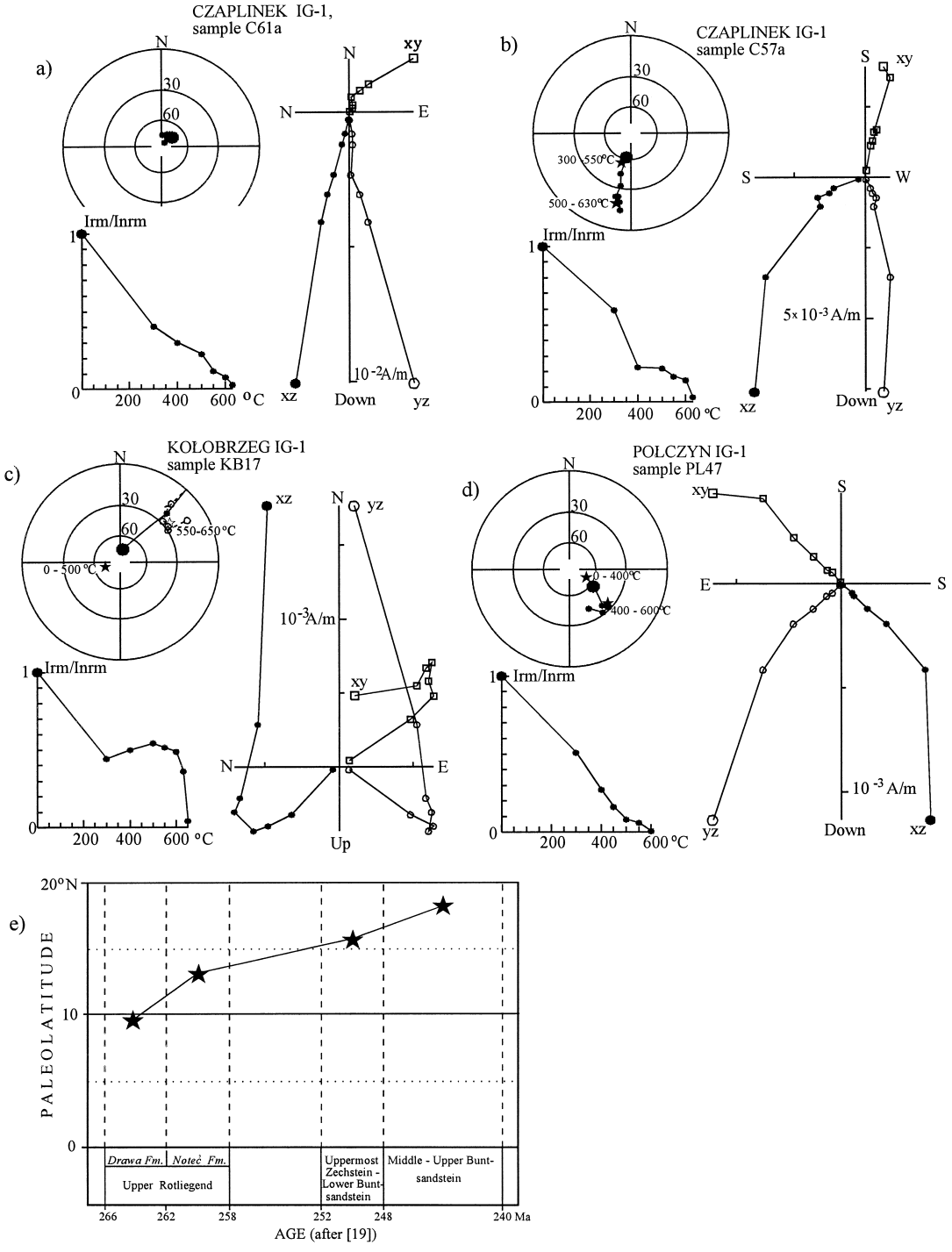
4.1.4. Lithuania: Karpenai quarry

NRM intensities of Early Zechstein carbonate rocks from Karpenai are one range lower than those observed in red beds. They never exceed 0.8 mA/m. During thermal demagnetization, at temperatures between 20°C and 300°C, a steep, most probably present-day, field component was removed (Fig. 4). Above 300°C a second normal polarity component with blocking temperatures higher than 400°C and an inclination about 30° was isolated. At that temperature the thermal demagnetization experiment was stopped due to the low intensities (below the sensitivity of the spinner magnetometer). Only one sample from the bottom part of the Karpenai profile revealed the presence of a shallow reversed component.

4.2. Samples from boreholes

About 50% of samples taken from the boreholes were completely remagnetized. During thermal de-

Fig. 5. Results of paleomagnetic investigations of Permian–Early Triassic rocks from unoriented bore cores sampled in the Polish Basin (stereographic projections of demagnetization paths, orthogonal plots and intensity decay curves). Bigger symbols mean NRM directions. Asterisks show fitted line directions. In (e) Late Permian–Early Triassic changes in paleolatitude of the Polish Basin calculated for an arbitrary locality (16°E; 52°N) are presented. Paleolatitudes were determined from the best quality boreholes results (see Table 2).


a)
CZAPLINEK IG-1,
sample C61a
N
30
60
Irm/Inrm
1
0
200 400 600 °C
N
xy
N
E
10 -2A/m
xz
Down
yz
b)
CZAPLINEK IG-1
sample C57a
300 - 550°C
500 - 630°C
Irm/Inrm
200 400 600 °C
S
xy
S
W
5x 10-3 A/m
xz
Down
yz
c)
KOŁOBRZEG IG-1
sample KB17
550-650 °C
0 - 500 °C
Irm/Inrm
200 400 600 °C
xz
N
yz
10-3A/m
xy
N
E
Up
d)
POŁCZYN IG-1
sample PL47
0 - 400°C
400 - 600°C
Irm/Inrm
200 400 600 °C
xy
S
E
S
10 -3 A/m
yz
Down
xz
e)
20°N
10
0
PALEOLATITUDE
Drawa Fm.
Noteć Fm.
Upper Rotliegend
Uppermost Zechstein - Lower Buntsandstein
Middle - Upper Buntsandstein
266 262 258 252 248 240 Ma
AGE (after [19])

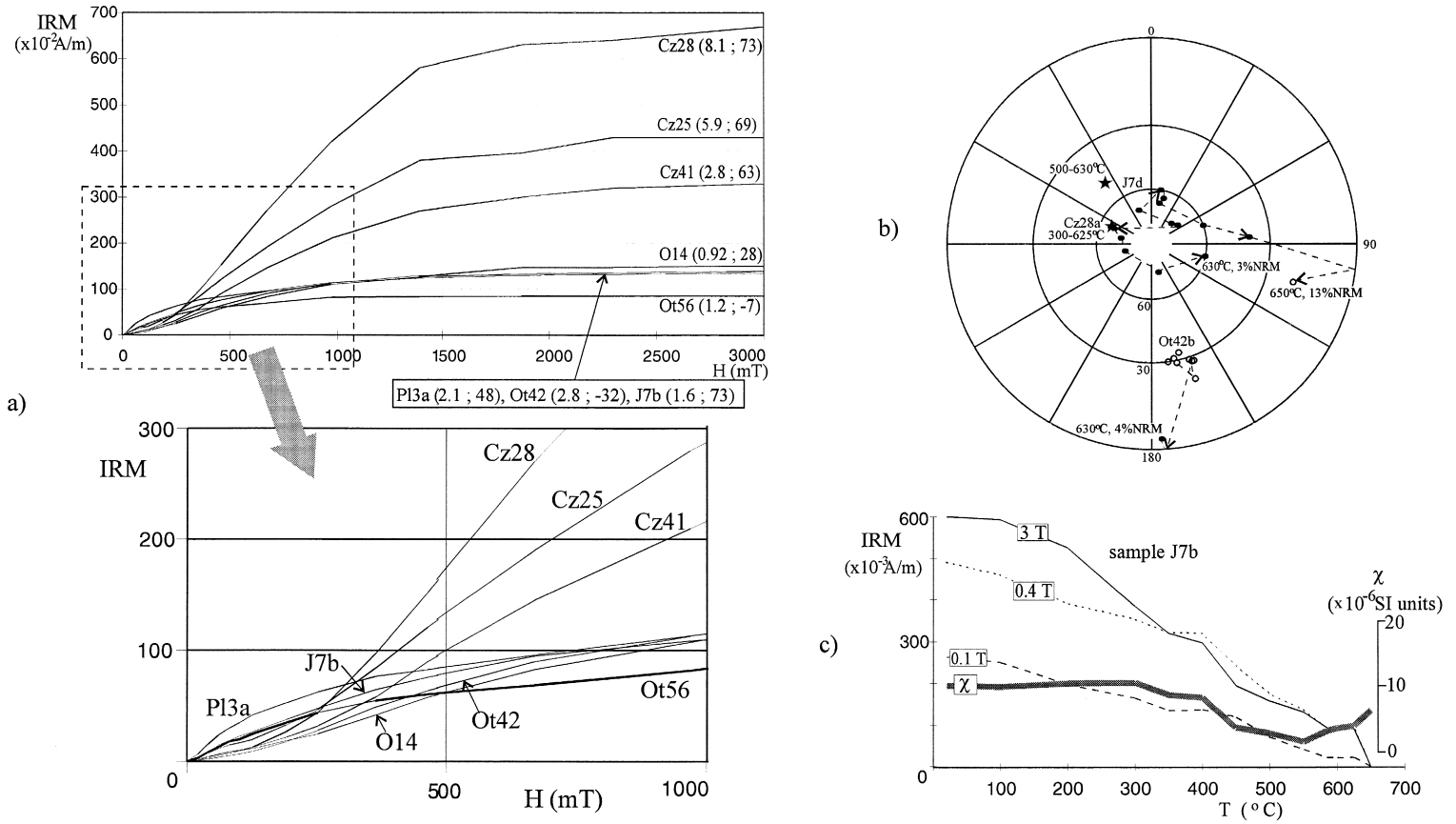


Fig. 6. (a) Isothermal remanent magnetization (IRM) curves obtained for red bed samples of Rotliegend (Cz25, Cz28, Cz41 and O14) and Buntsandstein (Ot42, Ot56, Pl3a and J7b) from unoriented boreholes of the Polish Basin. NRM intensities (in mA/m) and inclinations are listed. Samples Cz25, Cz28 and Cz41 were totally remagnetized along directions with steep inclinations. (b) Stereographic projection of demagnetization paths of some specimens coming from the same samples as in (a) (i.e. J7, Cz28 and Ot42). Asterisks mark fitted line directions. (c) Thermal demagnetization curves of orthogonal-axes IRM for Buntsandstein sample J7b and susceptibility curve (shaded line) obtained for this sample during subsequent heating.

magnetization up to 650°C they lost almost the whole NRM and showed only one component with steep inclination (Fig. 5a). Fortunately, in the other half of the sample set high-temperature components with shallow negative or positive inclinations were also present. They were often overprinted by a low–medium temperature component with a steep inclination and a similar or about 180° different (in the case when the high temperature component had negative inclination) declination (Fig. 5b,c). The maximum unblocking temperatures of shallow inclination components isolated from samples of the red beds were between 620°C and 660°C. The maximum unblocking temperatures of these components isolated in the samples, collected from thin horizons of grey sandstones or carbonates, were lower than 600°C (Fig. 5d). Mean inclinations of the characteristic high-temperature directions increase stepwise from 18° for Upper Rotliegend samples to 32° for Middle–Upper Buntsandstein samples (Table 2). The paleolatitude of the area investigated changed gradually from 9°N in the Late Permian to 18°N in the late Early Triassic (Fig. 5e).

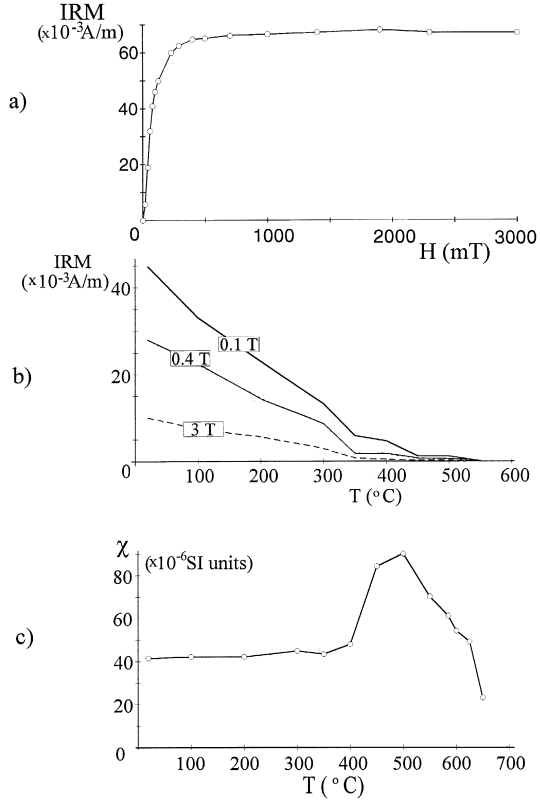


Fig. 7. (a) IRM curve, (b) thermal demagnetization of orthogonal-axes IRM curve and (c) susceptibility versus demagnetizing temperature curve obtained for a representative sample of Early Zechstein limestones from Karpenai quarry.

5. Magnetic carriers and anisotropy of magnetic susceptibility

Petrographic studies show that hematite is the most common ferric mineral in the Permian–Triassic red beds of the CEB. It is usually syndepositional, although a post-depositional origin is possible in the case of the Upper Rotliegend rocks, where partial, diagenetic, hematite cementation is observed [40].

The spectrum of unblocking temperatures noted in volcanites and red clastic sediments indicates that hematite is the main carrier of the characteristic components. However, a problem appears because a part of these rocks were strongly or even totally remagnetized in a direction approximately comparable with the present-day field direction. IRM acquisition curves of totally remagnetized samples differ from the curves obtained for unremagnetized or partly remagnetized samples. They have a different shape and higher IRM intensities in the fields above 400 mT (Fig. 6). It is likely that two generations of hematite occur in the red beds studied. This mineral,

Table 2
Summary statistics of inclinations derived from first and second category borehole samples from the Polish Permian–Triassic Basin

Age of rocks	Mean inclination	Standard deviation	Number of samples
Middle–Upper Buntsandstein	32	6	46, 46%N 54%R
Uppermost Zechstein–Lower Buntsandstein	29	5	81, 51%N 49%R
Upper Rotliegend–Noteć Fm.	24	6	27, 55%N 45%R
Upper Rotliegend–Drawa Fm.	18	6	18, 29%N 71%R

being the carrier of directions with the steep inclination, could be connected with recent or syn-Alpine fluid circulation (see e.g. [41]). In some samples, however, very strong, although not total, remagnetization could be caused by other processes. These strongly remagnetized samples may have similar IRM curves to the unremagnetized ones, but they are characterized by a strong decrease in susceptibility at temperatures between 400°C and 550°C (e.g. sample J7, Fig. 6). This phenomenon may be related to the presence of secondary maghemite, which converts to hematite during heating [42]. Values of maximum unblocking temperatures of grey clastic rocks and carbonates indicate that magnetite is here the main remanence carrier. In some carbonate samples, however, a contribution from ferric sulphides may not be excluded. The distinct decrease in isothermal remanent magnetization between 300°C and 350°C and slightly later increase in magnetic susceptibility could be indicative of the presence of these minerals (Fig. 7).

Minimum axes of low-field magnetic susceptibility anisotropy calculated for the sample set from the boreholes are generally vertical (Fig. 8). The ellipsoids of anisotropy are mostly oblate. The value of the foliation parameter is independent of the depth of sampling. Moreover, no connection between magnetic anisotropy and remagnetization of the samples

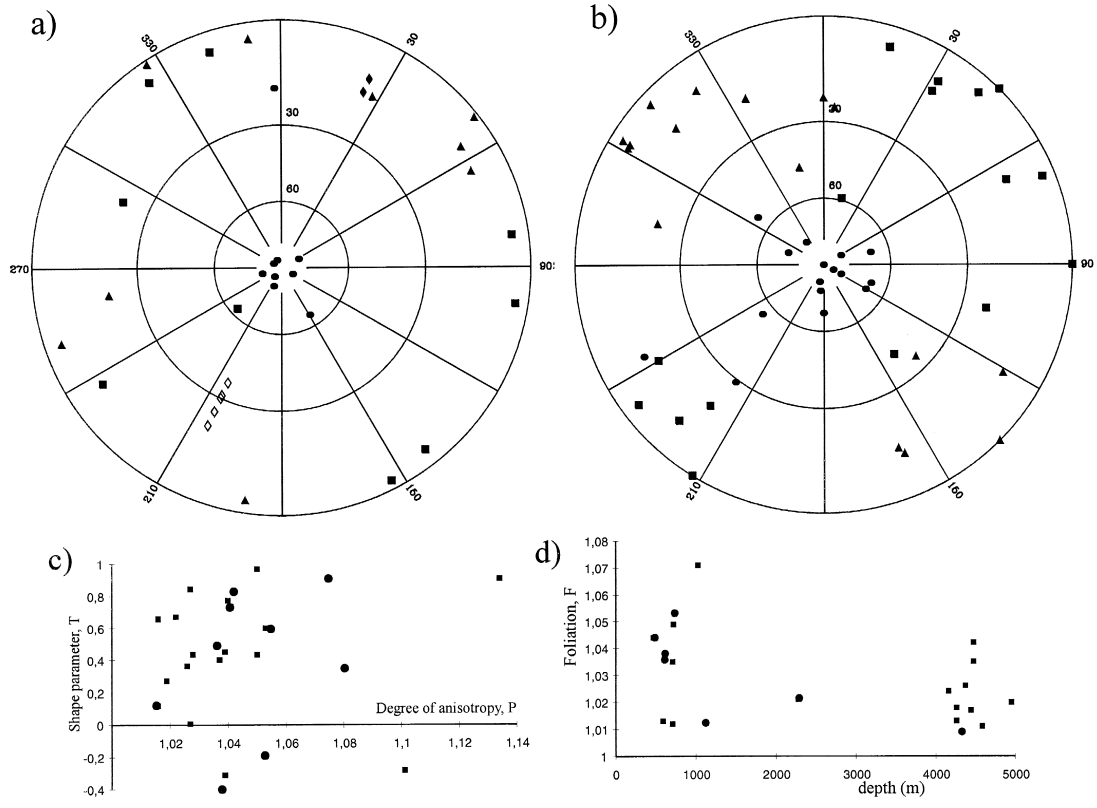


Fig. 8. (a) Stereographic projections of magnetic susceptibility anisotropy (AMS) axes calculated for some Upper Rotliegend and Buntsandstein specimens that contained well defined characteristic directions with shallow negative inclination. AMS axes are presented after rotation of characteristic directions (marked by diamonds) on meridian 205°. Squares = maximum axes; triangles = intermediate axes; circles = minimum axes. (b) Stereographic projection of AMS axes calculated for Upper Rotliegend and Buntsandstein specimens that were totally remagnetized (i.e., contained only directions with steep inclination). (c) Diagram of degree of AMS versus shape parameter and (d) diagram of foliation parameter versus depth of sampling constructed for both groups of specimens. In these figures the first group of specimens are marked by dots and the second group (specimens totally remagnetized) by squares.

studied is observed. Hence it is very probable that the observed foliation is not connected with the carriers of characteristic remanence.

The repetition of a comparable polarity pattern in several distant profiles (see next sections) and the lack of polarity changes in the places where magnetite became the main magnetic carrier (grey limestones and sandstones) instead of hematite, are the main paleomagnetic arguments supporting the suggestion of a primary origin for the characteristic components isolated in the present paper.

6. Magnetic stratigraphy

6.1. Outcrops

All the rocks studied are magnetized in the directions which provided the poles convergent with those expected for Permian–Early Triassic ones characteristic for Baltica (Fig. 9). The poles are located in the places which remain in agreement with the ages of the rocks studied. It should be stressed here that the pole isolated from the Early Zechstein Kupferschiefer deposits of southwest Poland [43] is also convergent with the Late Permian–Early Triassic segment of the Baltic apparent polar wander path. This pole was used as a cardinal argument for the Middle Triassic age of the Kupferschiefer mineralization in Poland [43]. However, as is clearly visible (Fig. 9), the position of this pole (latitude 49°N, longitude 157°E) does not prove that age of mineralization. The best defined Middle Triassic pole for stable Europe [44], with latitude 53.5°N and longitude 140.8°E, is separated from the discussed pole from Kupferschiefer deposits.

The Lower Rotliegend ('Autunian') rocks sampled in the Sudetes were only reversely magnetized. Zechstein samples from the Sudetes, Holy Cross Mts. and Lithuania were magnetized in both directions: normal and reversed. The lowermost Zechstein (Werra) sediments from the North Sudetic Basin and Karpenai showed magnetizations mostly in a normal direction. Only one sample from the bottom part of the Karpenai profile was reversely magnetized. The reversed polarity record was observed in most sediments up to the uppermost Zechstein (i.e. Top Terrigenous Series). The results from the outcrops show

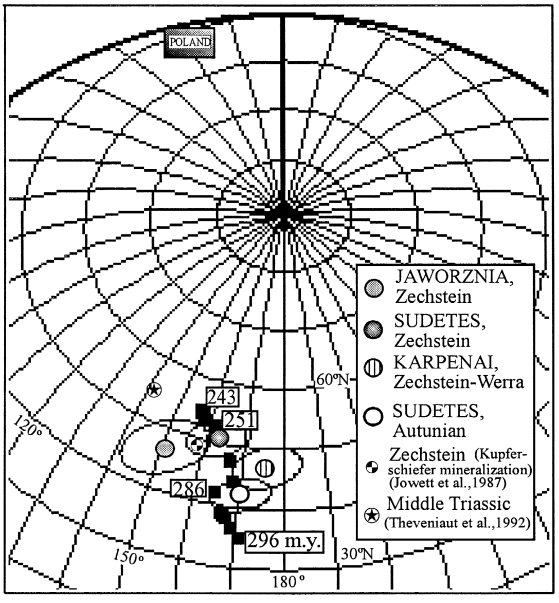


Fig. 9. Permian Apparent Polar Wander Path (APWP) for Baltica (smoothed with time resolution 2 Ma) and paleopoles from this paper. Pole of Jowet et al. [43] from Kupferschiefer (Zechstein–Werra) and Middle Triassic pole of Theveniaut et al. [44] are also presented. The APWP was constructed using GMAP package [49] and good quality poles ($Q > 3$) listed by Van der Voo [37]. Ages were adopted from the Permian time calibration of Menning [19].

that Permian–Early Triassic rocks are simply magnetized and they may also be magnetostratigraphically studied and interpreted from unoriented cores from boreholes located in the adjacent areas.

6.2. Boreholes

Only four samples of the Lower Rotliegend rocks taken from Objezierze IG-1 borehole give shallow inclination components. One of them, from the bottom part of the Lower Rotliegend, is normally magnetized (Fig. 10). The Upper Rotliegend rocks are characterized by a mixed polarity pattern. The first normal polarity zone occurs in the sediments of the Drawa Formation in the borehole Czaplinek IG-1. In the three boreholes investigated the normal polarity record is predominant in the upper part of the Noteć Formation. Within the entire Upper Rotliegend at least three normal and four reversed magnetic zones occur.

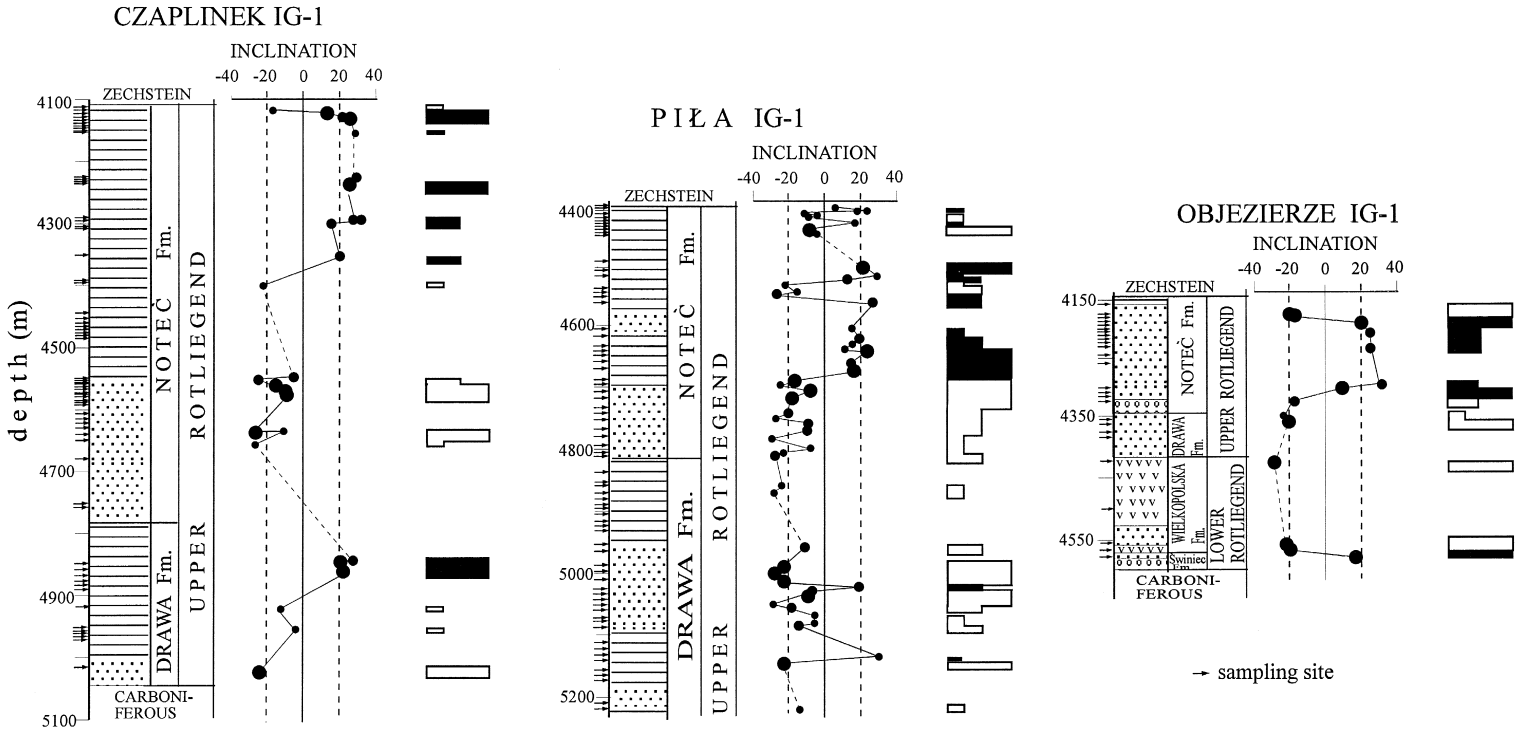


Fig. 10. Lithostratigraphy of Rotliegend sequences, characteristic inclinations of magnetization and interpreted polarity pattern noted in three boreholes from the Polish Basin. Black is normal polarity. The biggest dots and the longest polarity bars represent the first category of data (see text). Smaller dots and shorter bars are characteristic for the second category of data. Third category of data yielded inclinations and polarities marked by the smallest dots and shortest bars.

The magnetostratigraphic subdivision of the Zechstein–Buntsandstein sediments is based on more numerous samples than those used in the Rotliegend. The magnetozones were distinguished and numbered if the same polarity record was noted in more than one sample (Figs. 11 and 12). Samples from the lower Werra cyclothem PZ1 are characterized by normal polarity (borehole Jaworzna IG-1). This polarity is also dominant in the Werra sediments from the outcrops. Higher horizons of the Zechstein sequences, including PZ2, PZ3, lower parts of the Rewal Formation and the Top Terrigenous Series, are predominantly reversely magnetized. The sediments from Jaworznia quarry, allocated to the Top

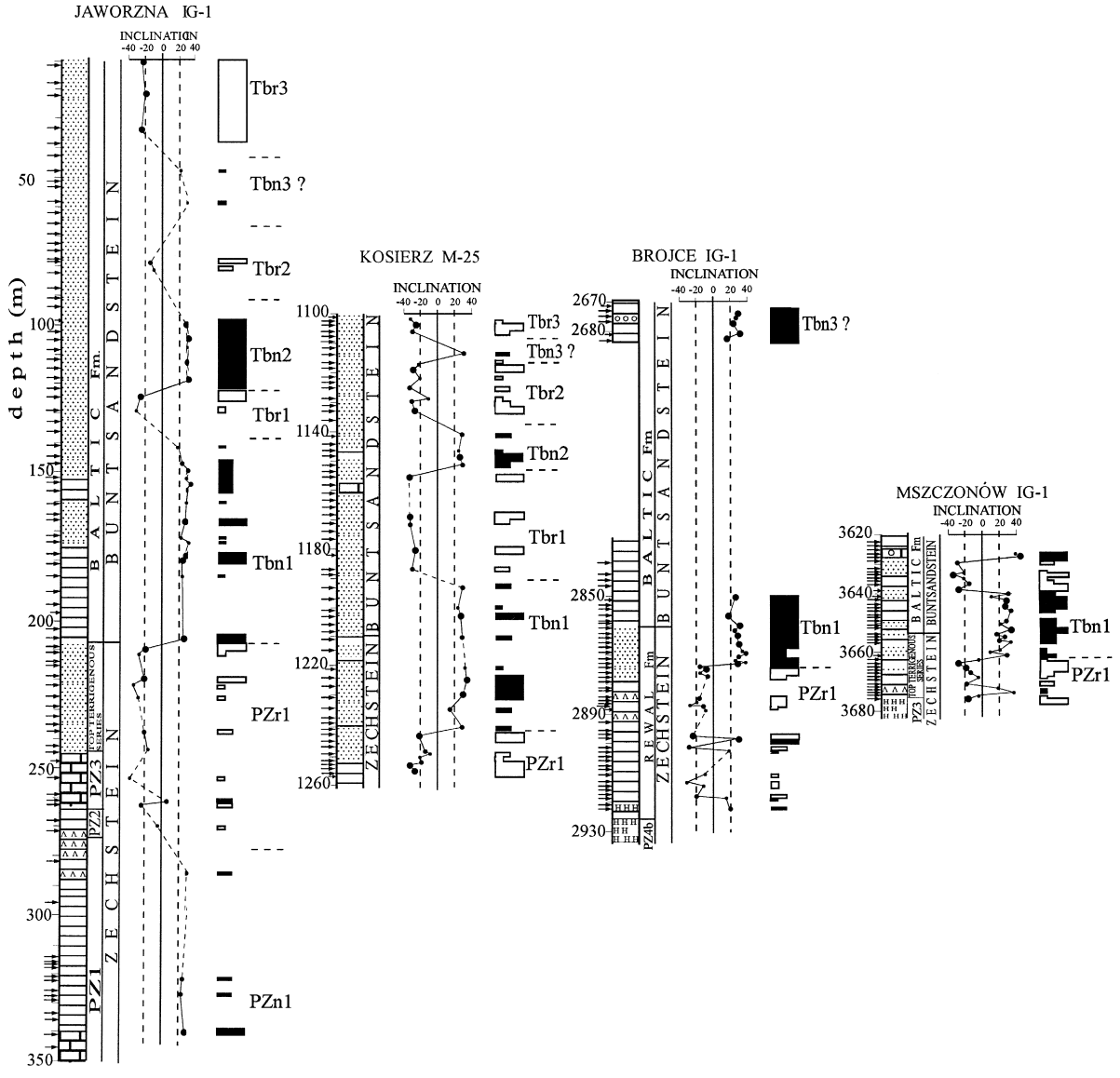
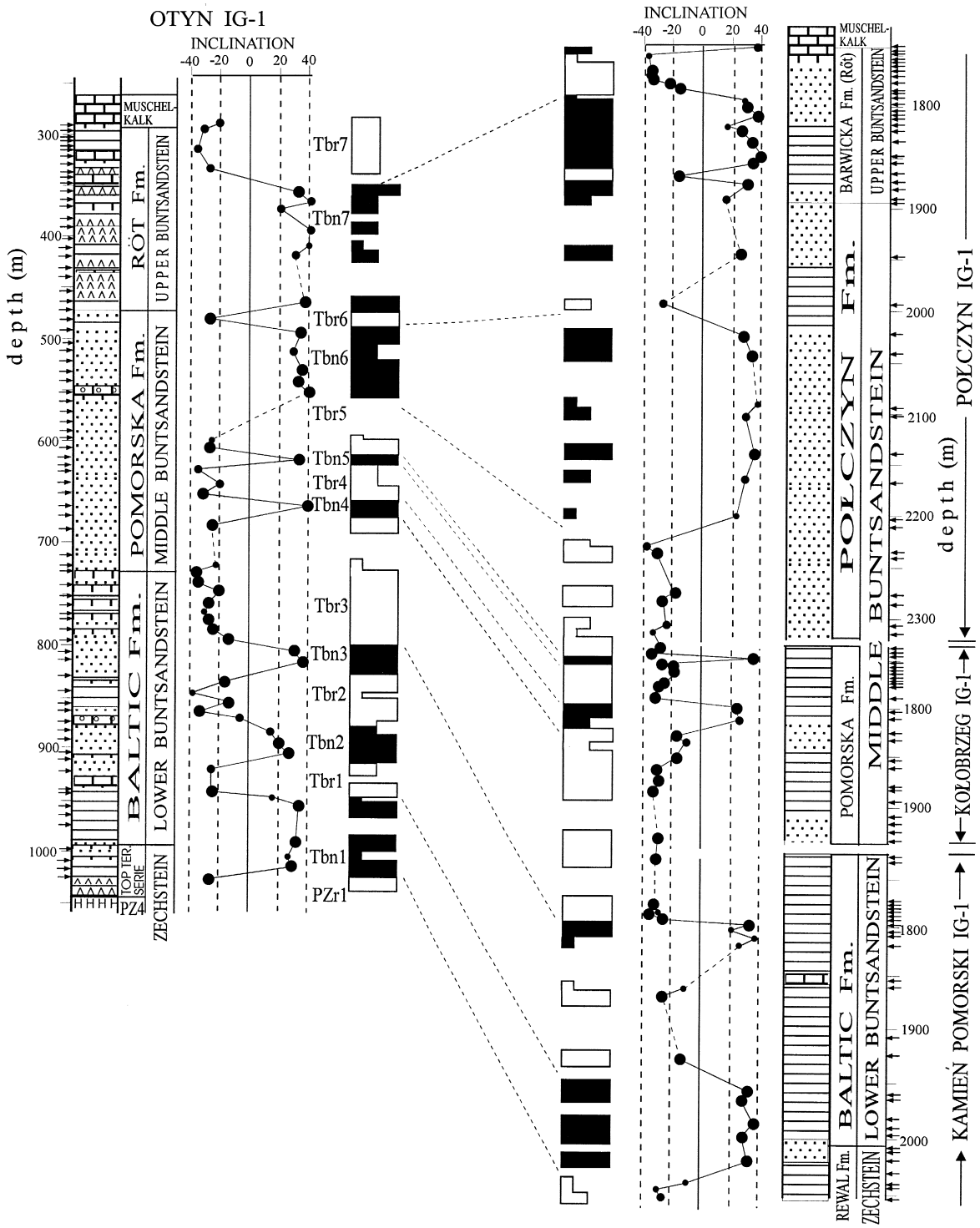


Fig. 11. Lithostratigraphy of Zechstein–lowermost Buntsandstein sequences, characteristic inclinations of magnetization and interpreted polarity pattern recorded in four boreholes from the Polish Basin. The polarity column is divided into numbered magnetozones.


OTYN IG-1
INCLINATION
-40 -20 0 20 40
depth (m)
MUSCHEL-KALK
RÖT Fm.
UPPER BUNTSANDSTEIN
POMORSKA Fm.
MIDDLE BUNTSANDSTEIN
BALTIC Fm.
LOWER BUNTSANDSTEIN
TOP TER. SERIE
PZ4
ZECHSTEIN
Tbr7
Tbn7
Tbr6
Tbn6
Tbr5
Tbn5
Tbr4
Tbn4
Tbr3
Tbn3
Tbr2
Tbn2
Tbr1
Tbn1
PZr1
INCLINATION
-40 -20 0 20 40
MUSCHEL-KALK
BARWICKA Fm. (Röt)
UPPER BUNTSANDSTEIN
POŁCZYN Fm.
MIDDLE BUNTSANDSTEIN
POMORSKA Fm.
REWAL Fm.
BALTIC Fm.
ZECHSTEIN
LOWER BUNTSANDSTEIN
depth (m)
KAMIEŃ POMORSKI IG-1
KOŁOBRZEG IG-1
POŁCZYN IG-1

Terrigenous Series, are also reversely magnetized. The total number of the Zechstein magnetozones seems to be at least eight.

A normal polarity zone was recorded near the Zechstein/Buntsandstein boundary in all profiles studied. In the Lower Buntsandstein and lower part of the Middle Buntsandstein at least five normal polarity zones occur, however, contradicting the rest of the Buntsandstein where the reverse polarity record is predominant (Fig. 12).

7. Magnetostratigraphic correlations

7.1. Composite polarity scale

The magnetic polarity pattern noted in the particular Permian–Buntsandstein formations is very similar. On this basis a composite polarity scale has been constructed (Fig. 13). This scale is not equally well documented since the Lower Rotliegend polarity scale is only approximate. It was established using the results from separate outcrops mainly from the upper part of Autunian. The majority of the second and third Zechstein cyclothems were not recognized magnetostratigraphically. The magnetozones were distinguished using the results of the second or first category of reliability. If the results were derived only from one sample, then the polarity columns are marked by shorter belts.

7.2. Illawarra Reversal

The end of the Kiaman superchron (i.e. Illawarra Reversal) is a first order magnetostratigraphic marker in the Permian. In the German part of the CEB, the end of the Kiaman was established in the lowermost formation (Schwerin Fm.) of the Upper Rotliegend with the estimated maximum age of 265 Ma [19]. The position of the first normal magnetozone in Polish profiles seems to be the same. It occurs in the lower formation of Upper Rotliegend (Fig. 13). Its correlation with the first post-Kiaman normal magnetozone in the global composite scale, however, is not unquestionable. It is not excluded that the end of Kiaman can be situated in the stratigraphic gap, with a duration of about 8 Ma [19], that separates the Lower and Upper Rotliegend rocks. The position of the Illawarra Reversal in the global composite scale is also controversial. The Russian platform data, that locate the Illawarra Reversal in the Tatarian [3,45], are inconsistent with the results from a continental red-bed section in north China where the end of Kiaman was interpreted in the Ufimian [46]. The discrepancy may be result of the incorrect stratigraphic allocation of the north China section [54].

7.3. Zechstein / Buntsandstein and Permian / Triassic boundaries

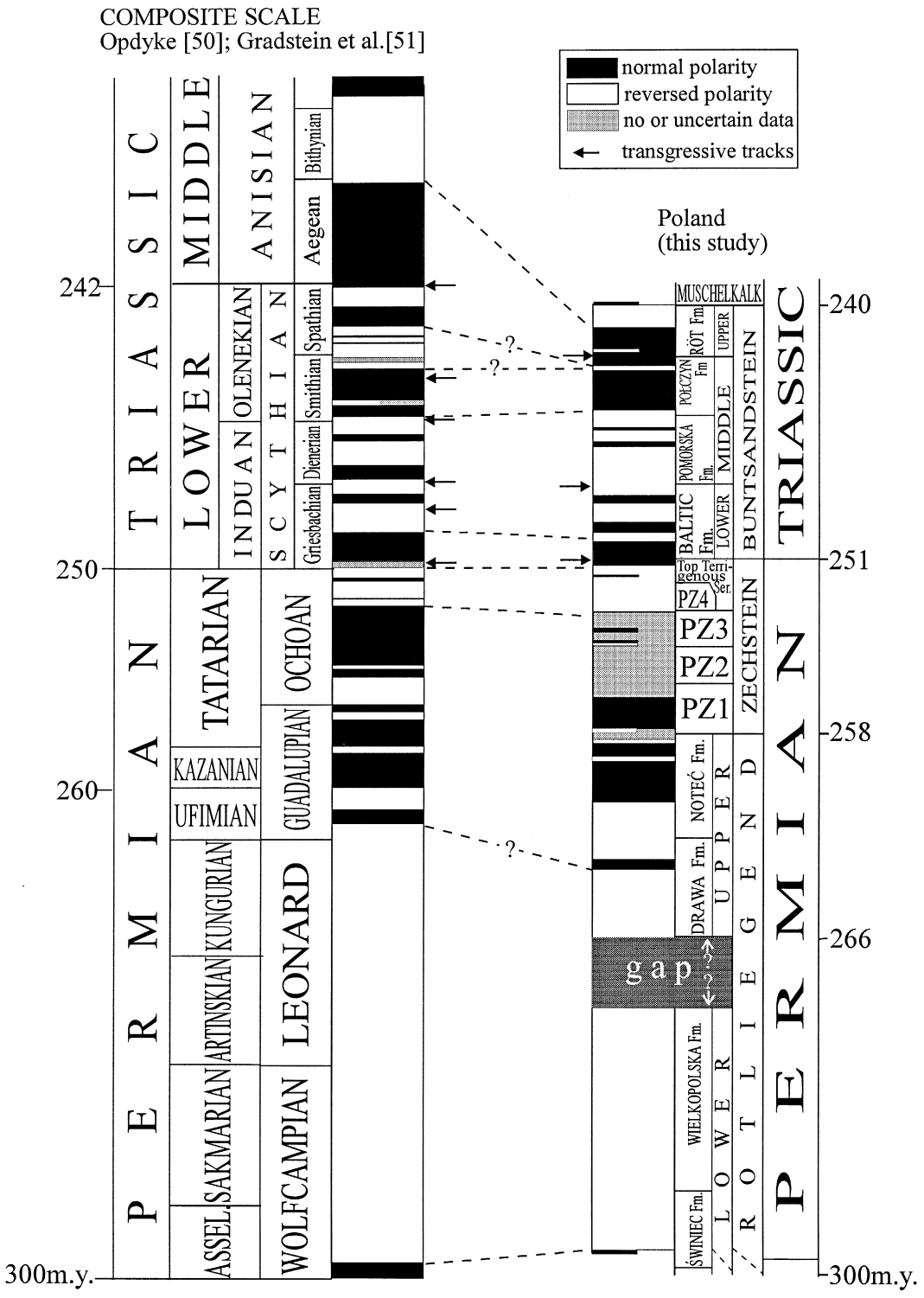
In the Polish part of the CEB the Zechstein/Buntsandstein boundary is defined as a point where first marine transgressive system tracks appear. They are regarded as being a result of the global Early Griesbachian transgression [27]. The bottom of the Baltic Formation contains Griesbachian microflora [33]. A Late Permian microflora was found in the bottom part of the underlying Top Terrigenous Series [22]. Hence it is very likely that the normal polarity magnetozone Tbn1 which occurs at the Zechstein/Buntsandstein boundary is the same as the magnetozone identified in the lowermost Griesbachian sequences in the Arctic area [7,8], China [9,11], and the Siberian traps [12]. The predominantly reversed polarity record that occurs in Top Terrigenous Series (magnetozone PZr1) is similar to the polarity pattern of the latest (Changxingian) Permian [11].

All facts presented above support the theory that the Zechstein/Buntsandstein boundary in the Polish part of the CEB is close to the Permian/Triassic one defined at the bottom of the Griesbachian stage.

7.4. Age of Buntsandstein formations

In the interval between the normal magnetozone Tbn1 of the lowermost Bundsandstein and the normal magnetozone Tbn6 of the upper part of Middle

Fig. 12. Lithostratigraphy of Buntsandstein sequences, characteristic inclinations and polarity pattern in four boreholes from the Polish Basin. Formations from Otyń IG-1 borehole, that was situated in the marginal part of the Polish Basin, were magnetostratigraphically correlated with adequate formations distinguished in three boreholes from its central part.


COMPOSITE SCALE
Opdyke [50]; Gradstein et al.[51]
normal polarity
reversed polarity
no or uncertain data
transgressive tracks
Poland
(this study)
242
250
260
300m.y.
240
251
258
266
300m.y.
TRIASSIC
PERMIAN
MIDDLE
LOWER
ANISIAN
SCYTHIAN
INDUAN
OLENEKIAN
Bithynian
Aegean
Spathian
Smithian
Dienerian
Griesbachian
TATARIAN
KAZANIAN
UFIMIAN
KUNGURIAN
ARTINSKIAN
SAKMARIAN
ASSEL
OCHOAN
GUADALUPIAN
LEONARD
WOLFCAMPIAN
MUSCHELKALK
RÖT Fm.
UPPER
POŁCZYN Fm
MIDDLE
POMORSKA Fm.
LOWER
BALTIC Fm.
BUNTSANDSTEIN
Top Terrigenous Ser.
PZ4
PZ3
PZ2
PZ1
ZECHSTEIN
NOTEĆ Fm.
DRAWA Fm.
UPPER
gap
WIELKOPOLSKA Fm.
LOWER
ŚWINIEC Fm.
ROTLIEGEND

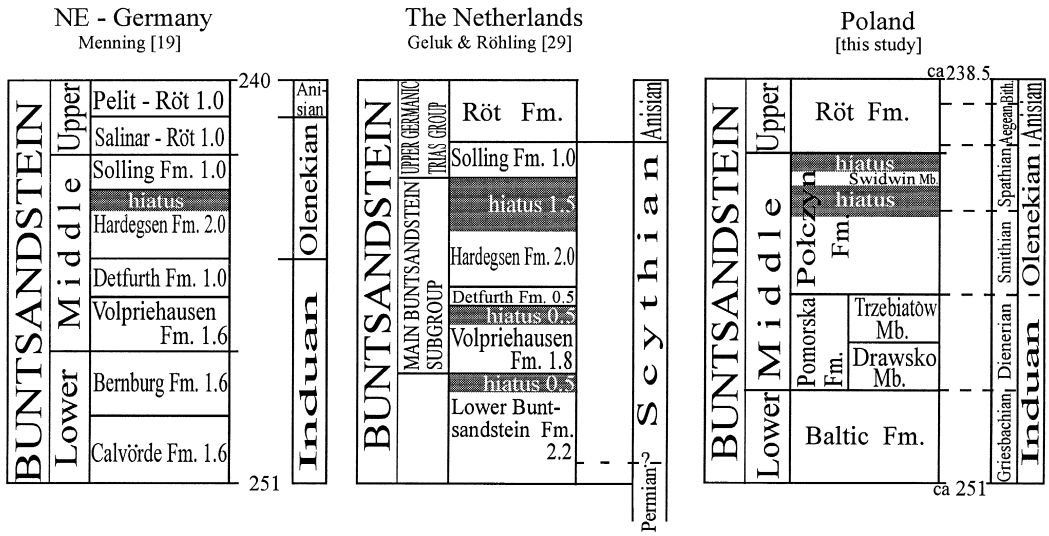


Fig. 14. Buntsandstein stratigraphy in the Central European Basin. Ages and estimated minimum duration of sequences are presented in Ma.

Bundsandstein a reverse polarity record predominates which is interrupted by four short normal polarity zones. Such a polarity pattern was also noted in the Arctic sections. The reversed polarity dominates there from the middle Griesbachian to the lowermost Smithian [7,8]. In that interval, 3 to 6 short normal magnetozones were confirmed. There is no geological evidence for a variation in the sedimentation rate in the part of Buntsandstein discussed, although the presence of some small gaps cannot be excluded. The correlation line placed at the bottom of the thick normal magnetozone Tbn6 (Figs. 12 and 13) seems very probable. Hence, Lower Buntsandstein and the lower part of Middle Buntsandstein could have been a time equivalent of the Griesbachian and Dienerian (Figs. 13 and 14). Lower Buntsandstein sediments in the Polish part of the CEB may be synchronous with the Griesbachian strata from Svalbard, where three normal magnetozones were also found [8].

In the Middle Buntsandstein sediments of distant sections Otyń IG-1 and Kołobrzeg IG-1, below magnetozone Tbn6, two short normal magnetozones, Tbn4 and Tbn5, occur (Fig. 12). This fact indicates that the Lower/Middle Buntsandstein boundary may be treated as relatively isochronous in the Polish part of the CEB. The magnetostratigraphic correlation (Fig. 12) also indicates that time equivalents of the Połczyn Formation are found in the upper part of the Pomorska Formation mapped in the more marginal part of the Polish Basin.

According to the results of magnetostratigraphic studies of the Arctic stratotypes, a notable part of Spathian rocks is magnetized in reversed direction [7,8]. In the Polish sections the upper parts of Middle Buntsandstein have a predominantly normal polarity record. Assuming an Anisian age of the Röt (see below), significant gaps or stratigraphic condensations near the Middle Buntsandstein/Röt boundary are very likely. Hiatuses could be located at the Middle Buntsandstein/Röt boundary and/or slightly deeper in the bottom of Swidwin Member (Figs. 13 and 14). In greater part of the Polish Basin the gap between Röt and underlying Middle Buntsandstein sediments is also postulated by geologists [55].

Normal polarity zone Tbn7 in the Lower Röt is most probably the same as the normal polarity zone characteristic of the latest Spathian–Early Anisian

Fig. 13. Composite Permian–Buntsandstein polarity scale for Poland (from [50,51]; ages after [19]) and its correlation with global Permian–Triassic composite scale. Transgressive tracks on a global scale are from [7,8].

sediments [14,47]. Because of the occurrence of reversed magnetozone Tbr7 in the upper part of the Röt, an alternative correlation could locate the whole Röt below the Anisian. However, such a location would not be in agreement with biostratigraphical data (e.g. [33]). Hence, either the majority or the entirety of the Röt sediments should be included in the Anisian.

8. Summary

The paleomagnetic studies of Permian–Early Triassic rocks sampled in the outcrops and unoriented boreholes from the Polish part of the Central European Basin allow creation of a composite magnetostratigraphic scale. Magnetostratigraphic correlation with the standard magnetostratigraphic scale shows that the Illawarra Reversal in the Polish sections is either located in the lower part of the Upper Rotliegend or might have not been recorded because of the stratigraphic gap postulated between the Lower and Upper Rotliegend formations. The magnetic polarity record and biostratigraphical and sedimentological evidence indicate that the Zechstein/Buntsandstein boundary marked in the Polish sections is close to the Permian/Triassic boundary defined as the bottom of the Griesbachian stage. The Griesbachian–Dienerian age of Lower Buntsandstein and the lower part of Middle Buntsandstein, as well as the early Anisian age of the whole or the majority of the Röt Formation seems to be supported and documented by the magnetostratigraphic correlation. Significant gaps or stratigraphic condensations near the Middle Buntsandstein/Röt boundary are very likely.

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